

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0408

Lone Tree 138 kV
December 26, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0408. The Interconnection Customer proposes a 327.0 MW wind facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on December 07, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 170 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$42,754,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Craven County Wood Energy Lp-Lone Tree 138.0 kV Line, the Osceola-Lone Tree 138.0 kV Line loads to 144.4% of its 217 MVA rating. Project DFAX: 100.0%.

Michigan Power Lp-Osceola 138.0 kV: 144.3% loading (214 MVA) following the outage of the Lone Tree Substation-Craven County Wood Energy Lp Substation 138.0 kV Line; DFAX = 100.0%.

For the loss of Lone Tree-Osceola 138.0 kV line, the Craven County Wood Energy Lp-Lone Tree 138.0 kV Line loads to 139.0% of its 226 MVA rating. Project DFAX: 100.0%.

Craven County Wood Energy Lp-Lowmoor 138.0 kV: 134.5% loading (230 MVA) following the outage of the Lone Tree-Osceola 138.0 kV; DFAX = 100.0%.

Grottoes-Weyers Cave 138.0 kV: 118.7% loading (258 MVA) following the outage of the Crandall-Dayon 345.0 kV L; DFAX = 5.1%.

For the loss of Louisa Generation Facility-Fluvanna P.S 345.0 kV, the Louisa Generation Facility 345/138kV Autotransformer loads to 102.0% of its 1,004 MVA rating. Project DFAX: 5.2%.

The Louisa Generation Facility-N Anna Substation 345.0 kV is overloaded to 101.4% of its 2,070 MVA emergency rating for the loss of the Carrabelle-Louisa Generation Facility 345.0 kV. The project contributes 10.0% DFAX (32.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Osceola-Lone Tree 138.0 kV	Rebuild line with higher-capacity conductor	\$3,312,000	100% (\$3,312,000)
Michigan Power Lp-Osceola 138.0 kV	Rebuild line with higher-capacity conductor	\$15,269,000	100% (\$15,269,000)
Craven County Wood Energy Lp-Lone Tree 138.0 kV	Rebuild line with higher-capacity conductor	\$13,447,000	100% (\$13,447,000)
Craven County Wood Energy Lp-Lowmoor 138.0 kV	Rebuild line with higher-capacity conductor	\$8,175,000	100% (\$8,175,000)
Weyers Cave-Grottos 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,223,000	20% (\$1,043,000)
Louisa Generation Facility 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$13,567,000	3% (\$364,000)
Louisa Generation Facility Substation-North Anna 345.0 kV	Reconductor line	\$27,867,000	4% (\$1,144,000)
	Total allocated to project		\$42,754,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.